

COP2800C Module 6 Graded Programming Assignment

In this assignment, we will use concepts covered in Module 6 by modifying the Module 5 Palmer Penguins analysis to incorporate user-defined methods. Our goal is to break down the program into smaller, reusable methods, improving readability and maintainability while incorporating best practices for method design.

Instructions

1. Complete the author and submission date lines in the ID header.
2. Refactor the program to use user-defined methods.

Modify the **PalmerPenguinsM5.java** program by implementing the following methods (**do not remove the TODO markers**)

Include a descriptive comment above each method definition and call.

- **TODO 1:** Create a method `readSpeciesData` that calls `CSVReader.readFile(FILE_NAME, 1)` and returns the resulting `String[]`. Replace the inline call to `CSVReader.readFile` with a call to this method in main.
 - **TODO 2:** Define a method `initializeSpeciesCount` that returns a new integer array of size `NUM_SPECIES`. Use this method to initialize `speciesCount` in main.
 - **TODO 3:** Implement a method `isEmpty` that takes a `String[]` array as a parameter and returns `true` if its length is 0, otherwise `false`. Use this method to check if `speciesData` is empty in main.
 - **TODO 4:** Write a method `countSpecies` that accepts `String[] speciesData` and `int[] speciesCount` as parameters, iterates through `speciesData`, and updates `speciesCount` accordingly. Replace the inline counting logic with a call to this method.
 - **TODO 5:** Implement a method `printSpeciesCount` that prints the count of each species using the predefined constants. Replace the three `System.out.println` statements in main with a call to this method.
3. Test Your Program
 - Run your program and verify that it correctly outputs the species counts.
 - Ensure that each method is correctly implemented and called from main.

Example output (unchanged from Module 5):

Chinstrap count = 68

Gentoo count = 123

Adelie count = 151

4. **Screenshot Your Output**
 - Take a screenshot of your program's output in the "Run I/O" pane of the jGrasp IDE.

- **DO NOT** include the source code in the screenshot.

5. **Submission**

Submit your completed **PalmerPenguinsM6.java** file to the GitHub repository.

- Do not include .class files.
- Ensure your code follows our Java coding conventions (indentation, comments, line length).

Upload your screenshot to the repository.